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AI-Driven Probabilistic Approach for Structural Uncertainty Quantification Unveiled In UAE

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Abstract

Recent advancements in geoscience have underscored the transformative potential of artificial intelligence (AI) in addressing the persistent challenges associated with subsurface structural characterization. Structural uncertainty, defined as the inherent variability or error in delineating subsurface geological features such as reservoir geometry and spatial positioning, remains a critical concern for reliable geophysical modeling. Uncertainty is introduced by factors such as seismic data quality, variations in seismic velocities, subsurface anisotropy, the limited and noisy nature of well data, and environmental or operational influences that may compromise data integrity. These uncertainties have far-reaching implications, extending beyond geoscientific risk to directly influence project economics and operational safety. For example, imprecise depth measurements can result in suboptimal well placement, missed reservoir targets, inappropriate design of casing and completion programs, increased development costs and delays, as well as greater exposure to drilling hazards such as over-pressure zones or unexpected water influx. Accurate quantification of this uncertainty is essential, as it directly influences the credibility of subsurface reservoir models, which are fundamentally constructed from incomplete or noisy geophysical datasets. Emergence of AI-driven probabilistic frameworks has significantly enhanced the ability of geoscientists to quantify structural uncertainty throughout the modeling workflow. These advanced methodologies are reshaping industry practices in subsurface exploration, reservoir development, resource management, and risk mitigation, particularly within geologically complex settings.

Within this evolving landscape, AI technologies, particularly deep learning models such as convolutional neural networks (CNNs) are increasingly integrated into seismic interpretation and structural modeling. These data-driven algorithms exhibit superior capabilities in pattern recognition and spatial classification, automating the extraction of geological features from seismic datasets. The implementation of AI-based tools in the workflow facilitates the rapid generation of subsurface models, reduces interpreter bias, and standardizes the analysis of seismic data across varying vintages and reprocessing versions. As a result, the reproducibility and reliability of structural interpretations are markedly improved. Moreover, these approaches support comprehensive validation of experimental workflows, including depth uncertainty mapping via blind well tests. Traditional deterministic modeling, which relies on single-point estimates

and manual interpretation, is limited in its ability to capture geological ambiguity and is sensitive to errors introduced at multiple stages of data acquisition and processing. This limitation often leads to the propagation of uncertainties, impacting the robustness of volumetric and structural predictions. In contrast, probabilistic workflows are specifically designed to sample the plausible spectrum of geological scenarios, providing statistically robust confidence intervals for resource estimates. The adoption of these approaches enhances the ability of asset teams to design optimized field development strategies and manage subsurface risks more effectively by quantifying and communicating the full range of outcome uncertainties.

This paper highlights the development of specialized AI frameworks within the geosciences that extend beyond conventional machine learning applications. Distinct from traditional AI, these advanced techniques are engineered to estimate subsurface geological properties, such as reservoir depth, by minimizing discrepancies between model outputs and observed geological data, which are often limited to sparse well control points. These frameworks utilize probabilistic modeling and statistical rigor, aiming to minimize estimation bias and variance. Uncertainty quantification is achieved through the calculation of the standard deviation of depth errors, providing a robust metric for model validation. Unlike deterministic approaches, these specialized AI-driven methodologies employ mathematically objective models, resulting in more consistent and unbiased uncertainty assessments. A practical application of this AI-driven methodology was demonstrated in the carbonate reservoirs of Field-A. Results indicated a remarkable reduction in project turnaround times, with deliverables produced in days rather than the months required by conventional manual workflows, representing a time saving of up to 70%. This approach not only accelerates decision-making in the early phases of oil recovery and field development but also delivers consistency and scalability across multiple projects by standardizing the analytical process. The system is designed for continuous learning, integrating new data to further refine its predictive capabilities over time.

Introduction

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the reproducibility and reliability of structural interpretations are markedly improved. Moreover, these approaches support comprehensive validation of experimental workflows, including depth uncertainty mapping via blind well tests. Traditional deterministic modeling, which relies on single-point estimates and manual interpretation, is limited in its ability to capture geological ambiguity and is sensitive to errors introduced at multiple stages of data acquisition and processing. This limitation often leads to the propagation of uncertainties, impacting the robustness of volumetric and structural predictions. In contrast, probabilistic workflows are specifically designed to sample the plausible spectrum of geological scenarios, providing statistically robust confidence intervals for resource estimates. The adoption of these approaches enhances the ability of asset teams to design optimized field development strategies and manage subsurface risks more effectively by quantifying and communicating the full range of outcome uncertainties.

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In conclusion, the integration of AI and probabilistic methods marks a paradigm shift in the quantification of structural uncertainty for geoscientific applications. By moving beyond deterministic paradigms, these innovative workflows deliver improved fidelity in subsurface modeling, enhanced project efficiency and enable more informed decision-making throughout the reservoir lifecycle. The adoption of such methodologies is poised to become an industry standard, offering significant benefits for resource evaluation, asset development, and risk management in the oil and gas sector.

Approach

AI Driven Probabilistic Geomodelling

The theoretical background of probabilistic geomodelling is found in the ‘Theory of Regionalized Variable’ (G. Matheron 1970). The theory considers that the (unknown) subsurface property may be expressed as the unique outcome of a Random Function Z , defined at any location (x) in space, as a random variable $Z(x)$:

$$Z(x) = m(x) + R(x)$$

Where:

- $Z(x)$ represents the true (unknown) subsurface property at location (x)
- $m(x)$ is the average value of $Z(x)$ at location(x)
- $R(x)=Z(x)-m(x)$ i.e. the $Z(x)$ residual capturing the spatially correlated random component around the mean.

When based on scattered direct measurements (e.g., well data), and field-wide indirect observations (e.g., seismic data), it is not possible to fully determine $Z(x)$ at every location. However, by computing experimental spatial statistics from these datasets, one can:

- Estimate $m(x)$ the spatial trend (mean behavior), and
- Determine $C(x,y) = \text{Cov}(x)(R(x), R(y))$, the spatial covariance function describing how residuals vary as a function of distance and direction.

Characterizing and specifying these two components $m(x)$ and $C(x,y)$ from the data is enough to provide a mathematical foundation for optimal spatial estimation of $Z(x)$, known as "kriging". Kriging is the probabilistic model operator used to derive the best linear unbiased estimate of Z at a target location (x_0).

The kriging estimator is defined as: $Z_K^*(x_0) = m^*(x_0) + \sum_{i=1}^N \lambda_i R(x_i)$

With minimized estimation variance $\sigma_K^2(x_0) = \sum_{i=1}^N \lambda_i * \text{Cov}(R(x_i)R(x_0))$

Where:

- $m^*(x_0)$ represents the kriging estimate of the average value of $Z_K^*(x_0)$ at location (x_0)
- $\sum_{i=1}^N \lambda_i R(x_i)$ represents the linear combination of N neighboring residual values at locations $x_{(i)}$
- $\lambda_{(i)}$ are kriging weights applied to residual $R(x_{(i)})$ that minimize estimation variance.
- $\sigma_K^2(x_0)$ is the kriging variance that is the variance of the estimation error minimized by the kriging operator.

Kriging estimates satisfy two key statistical properties:

- Unbiasedness: $E\{Z(x_0) - Z_K^*(x_0)\} = 0$, i.e. on average, the kriging estimation error is equal to zero
- Minimum Estimation Variance: $\sigma_K^2(x_0) = \text{Var}\{Z(x_0) - Z_K^*(x_0)\}$ minimized

Kriging is recognized as the optimal linear estimator in geostatistics, providing the lowest achievable estimation uncertainty. It delivers the most probable value (P-50) for subsurface properties at each spatial location and quantifies associated uncertainty through the kriging standard deviation. This enables the definition of confidence intervals, such as P-10 and P-90 bounds, which are instrumental in decision-making processes for well placement and subsurface resource evaluation.

The kriging approach generates a spatially continuous estimate that is compared directly with the unknown true property values across the model domain. Its role as a foundational technique in probabilistic geomodelling makes it indispensable for minimizing uncertainty in structural modeling, depth conversion, and volumetric calculations. In terms of methodology, the kriging algorithm is analogous to convolutional neural networks (CNNs) in artificial intelligence, with the key distinction that kriging weights are derived analytically from spatial covariance models rather than learned through data-driven training.

Operating within a formal probabilistic framework, kriging relies on well-defined assumptions about both the spatial trend (mean) and the spatial covariance of residuals around that trend. It is tailored for the optimal estimation of additive variables and their linear transformations, including averages, convolutions, and gradients. As a local estimator, kriging focuses on providing accurate and smooth representations of subsurface properties, producing P-50 realizations that avoid artificial extrema unless warranted by proximal data. However, kriging is not designed for non-linear transformations, including volumetric computations above contact cutoffs or for capturing complex geometries and connectivity patterns.

To overcome these limitations, conditional spatial simulations are implemented within the same theoretical framework. Unlike kriging, which aims for accuracy, spatial simulations are designed to enhance realism by generating multiple equiprobable realizations of the random function $Z(x)$. These realizations honor both the mean, and the spatial covariance structure used in kriging, resulting in models that fluctuate around the kriging-based P-50 scenario, typically within the bounds defined by ± 2 standard deviations. Since kriging performs exact interpolation, all simulations precisely match input data at control points.

A critical aspect of kriging is that its standard deviation encapsulates uncertainty in both the trend and the residual components. As a result, conditional simulations that incorporate both sources of uncertainty provide a comprehensive representation of structural variability. This contrasts with standard conditional spatial simulations, which often focus solely on residual variability and may underrepresent the uncertainty inherent in trend estimation, highlighting a practical distinction.

From an artificial intelligence perspective, conditional spatial simulation algorithms facilitate systematic exploration of the plausible range of subsurface scenarios, offering a robust synthetic dataset for training advanced machine learning models, including CNNs. These capabilities are foundational to contemporary probabilistic modeling workflows, supporting rigorous quantification and management of structural uncertainty in geoscientific applications.

Probabilistic Models – Quantifying Structural Uncertainty

Under this theoretical framework, various probabilistic models have been created to tackle specific estimation challenges encountered at each step of building the structural model:

- Data qualification
- Velocity modeling
- Depth conversion
- Volumetrics and probability maps
- P-10 P-50 P-90 structural cases

Each model is defined by explicit, testable assumptions regarding the spatial behavior of both the mean and the spatial covariance of the property being estimated, ensuring full traceability and reproducibility. These models are designed to provide the most accurate possible estimation while delivering a quantifiable measure of the minimized spatial uncertainty. Three models are involved in structural geomodelling:

- Factorial Kriging: for spatial data conditioning (data qualification)
- Multivariate Bayesian Cokriging: for layer velocity modeling and depth conversion
- Multivariate Bayesian Co-Simulations: for volumetrics probability maps and P-10 P-50 P-90 structural cases

The factorial kriging model splits a random function $Z(x)$ as the sum of independent spatial components and looks for estimating one of them based on sampled $Z(x_{(i)})$ (*Reference: Geostatistical Seismic Data Processing: Equivalent to Classical Seismic Processing + Uncertainties! A. Shtuka 2011*). It is used in our case study for best estimating the spatial seismic signal $Z_s(x)$ from the measured seismic data $Z_m(x)$ that contains both signal $Z_s(x)$ and noise $Z_n(x)$.

$$Z_m(x) = Z_s(x) + Z_n(x)$$

When applied to seismic amplitude pre stack or post stack data, the probability model considers the measured raw seismic amplitude data $A(x, t)$ as a realization of a random function $Am(x, t)$ defined in

space and time. This random function (RF) $Am(x, t)$ can be decomposed in different components (factors) corresponding to the true unknown signal amplitude $A(x, t)$ and the noise $N(x, t)$:

$$Am(x, t) = A(x, t) + N(x, t)$$

Assuming that:

- Raw amplitudes are normalized along the time axis with 0 average
- Noise is not correlated with the signal

$Cam(hx, ht)$, the spatial covariance function of $Am(x, t)$ is the sum of the spatial covariance functions of the signal $A(x, t)$ and noise $N(x, t)$:

$$Cam(hx, ht) = Ca(hx, ht) + Cn(hx, ht)$$

Within this probabilistic model, the kriging operator is called factorial (co) kriging as it aims at best estimating the factors of the decomposition of the measured amplitude data by co-kriging them from the raw amplitude measured data:

$$A_K^*(x_0, t_0) = 0 + \sum_{i=1}^N \lambda_i Am(x_i, t_i)$$

$$\text{With } \sigma_K^2(x_0, t_0) = \sum_{i=1}^n \lambda_i^* \text{Cov}(A(x_i, t_i) A(x_0, t_0))$$

Where:

- 0 is the average value of the normalized amplitude data
- $\sum_{i=1}^N \lambda_i Am(x_i, t_i)$ represents the linear combination of N neighboring measured amplitude values at locations $x_{(i)}$
- $\lambda_{(i)}$ are co kriging weights applied to raw amplitudes $Am(x_{(i)}, t_{(i)})$ that minimize estimation variance attached to the kriging estimation of the amplitude signal $A_K^*(x_0, t_0)$.
- $\sigma_K^2(x_0, t_0)$ is the kriging variance that only involves the Covariance function of the amplitude signal and not of the covariance function of the measured amplitude or of the noise.

The factorial kriging model acts as a geophysical filter. For seismic amplitudes, depending on the chosen covariance model, the decomposition of measured amplitudes into "signal" and "noise" components produce results like those from well-known deterministic filters like the median filter, Wiener filter, or F-K filter.

For seismic velocities, the factorial kriging model separates a seismic layer interval velocity into its "geological" spatial component and a local residual geophysical velocity artifact, which is not correlated to the time isopach, meaning it is not "geological" (*Reference: Are PSDM Depth Interpretations Reliable? L. Sandjiv 2016*).

$$Z_{SD}(x) = Z_{SDT}(x) + Z_{SDRA}(x)$$

Where:

- $Z_{SD}(x)$ is the Seismic depth product of seismic time by seismic velocity at location (x)
- $Z_{SDT}(x)$ Seismic depth correlated to seismic time that is consistent with geology.

$$Z_{SDT}(x) = \sum_i a_i^* f_i(Tx)$$

$Z_{SDT}(x)$ is expressed as the product of a consistent geophysical velocity law $V(x)$ by the seismic time or isopach. $V(x)$ itself is expressed as a linear combination of spatial functions of time.

- $Z_{SDRA}(x)$ Seismic depth residual not correlated to seismic time considered as a seismic artefact

Using the final decomposition of $Z_{SDT}(x)$ and an appropriate choice of covariance model for stationary $Z_{SDRA}(x)$

$$Z_{SD}(x) = \sum_i a_i f_i(Tx) + Z_{SDRA}(x)$$

Factorial (co) kriging algorithm enables us to best estimate the seismic depth artefact $Z_{SDRA}(x)$ from known seismic depth $Z_{SD}(x)$ and seismic time $T(x)$

$$Z_{SDRA}^*(x) = \sum_{\alpha} \lambda_{\alpha} Z_{SD}(x_{\alpha}) + \sum_{\beta} \mu_{\beta} a_i f_i(Tx)$$

With

$$\sigma_K^2 Z_{SDRA}(x_0, t_0) = \sum_{\alpha} \lambda_{\alpha} * \text{cov} \left(Z_{SDRA}(x_i, t_i) Z_{SDRA}(x_0, t_0) \right) + \sum_{\beta} \mu_{\beta} a_i f_i T(x)$$

To our knowledge there is no such geophysical filter equivalent to factorial kriging for the spatial conditioning of seismic velocity. This is likely because the "optimal" seismic velocity field for migration and imaging does not necessarily align with the one suited for geological velocity modeling. Factorial kriging not only extracts the spatial component of seismic velocities that positively contributes to velocity modeling but also quantifies the associated uncertainty through the estimation standard deviation. This represents a valuable complement to pre-stack depth migration techniques, helping to ensure the spatial consistency of depth interpretations while providing a meaningful assessment of their uncertainty.

The multivariate Bayesian co-kriging model is renowned for its application in multilayer velocity modeling and depth conversion, as illustrated in [Figure-1](#) (Reference: *Depth conversion and associated uncertainties using consistent velocity model: A probabilistic unified model based on Bayesian approach*. L. Sandjivy 2009).

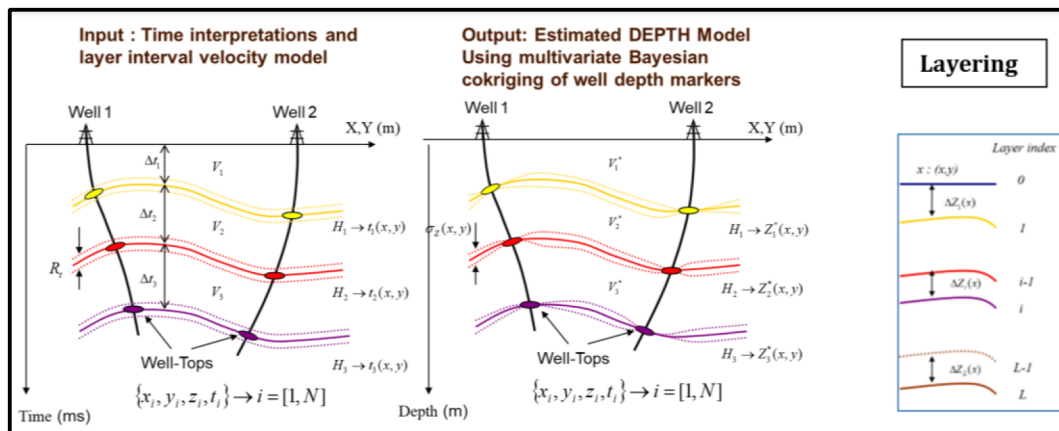


Figure 1—Multivariate Bayesian co kriging model input output and layering (courtesy Seisquare Arben Shtuka)

The depth of the top layer 'L' is expressed as: $Z_L(x) = \sum_{i=1}^L \Delta Z_i(x)$

- $Z_{(L)}(x)$ Random function representing the depth of horizon $H_{(L)}$

- $\Delta Z_{(i)}(x)$ Random function representing the geological isopach of layer i at horizon $H_{(i)}$

Each layer depth isopach is expressed as: $\Delta Z_i(x) = V_i(x) \Delta T_i(x)$

- $V_{(i)}(x)$: Random function representing the geological interval velocity of layer
- $\Delta T_{(i)}(x)$ Random function representing the seismic vertical arrival time isopach of layer i

Both travel time $T_i(x)$ and velocity $V_i(x)$ are themselves considered as random functions, each one expressed as the sum of mean and a residual random function:

$$\begin{aligned} T_i(x) &= t_i(x) + R_i^t(x) \\ V_i(x) &= \bar{V}_i(x) + R_i^v(x) \end{aligned}$$

- $\bar{V}_i(x)$: average interval velocity of layer i modeled by a velocity law
- $t_{(i)}(x)$: interpreted single travel time corresponding to the horizon $H_{(i)}$
- $R_i^v(x)$: interval velocity residual random function of layer i not captured by the velocity law
- $R_i^t(x)$: time residual random function representing picking uncertainty of the horizon $H_{(i)}$

The depth of horizon L depth, called Z_L , can be developed as such:

$$Z_L(x) \approx \underbrace{\sum_{i=1}^L \bar{V}_i(x) \cdot \Delta t_i(x)}_{\text{Trend}} + \underbrace{\sum_{i=1}^L R_i^v(x) \cdot \Delta t_i(x)}_{\text{residual due to the velocity uncertainty}} + \underbrace{\bar{V}_L(x) \cdot R_L^t(x)}_{\text{residual due to the time uncertainty}}$$

The horizon depth is known at marker depth location $Z_i(x_0)$ for each layer i , including layer L , x_0 sampled points being possibly different for each layer i . The trend $m_L(x) = \sum_{i=1}^L \bar{V}_i(x) \cdot \Delta t_i(x)$ at layer L corresponds to the average isopaches of all the layers above layer L . The spatial covariance of Z_i may be inferred from the above as each single average isopach is the product of the seismic vertical isopach time $\Delta t_i(x)$ by the interval velocity law $(V_i)(x)$.

It is convenient to linearize the velocity laws as functions of the seismic arrival time, using transfer functions. This enables us to consider the velocity law parameters (constant V_0 , rescaling factor A , V_0 and k parameters) as random variables with a priori acceptable range of values defined by the average or mean of the distribution being the joint covariance matrix being $\text{Cov}(B_i, B_k) = \sigma_{ik}$.

The spatial covariance $C(x, y) = \text{Cov}(Z_L(x), Z_L(y))$ between $R_L^v(x)$ and $R_L^v(y)$ is a linear combination of the spatial covariance of time and interval velocity residuals at all layers, and is derived as follows:

$$\begin{aligned} \text{Cov}(R_L^v(x), R_L^v(y)) &= \sum_{m=1}^L \underbrace{\Delta t_m(x) \Delta t_m(y)}_{\substack{\text{Spatial covariance of} \\ \text{Depth residuals for layer } L}} * \underbrace{\text{Cov}(R_i^v(x), R_i^v(y))}_{\substack{\text{Spatial covariance of} \\ \text{Residual velocity for all layer above } L}} + \underbrace{\bar{V}_L(x) \bar{V}_L(y)}_{\substack{\text{Spatial covariance of} \\ \text{Residual time for layer } L}} * \text{Cov}(R_L^t(x), R_L^t(y)) \end{aligned}$$

Once $m(x)$ and $C(x, y)$ have been characterized, the probabilistic depth conversion workflow runs on a specific kriging algorithm for computing the kriged estimate of the depth property: $Z^{(k*)}(x)$. The specific kriging algorithm that corresponds to the multilayer vertical depth conversion workflow is called co-Kriging depth with seismic time as an external drift and under Bayesian velocity constraints on the drift coefficients.

In mathematical terms, $Z_{(k^*)}(x)$ is the linear combination $\sum_{i=1}^N \lambda_{k_i} \cdot Z_R(x_i)$ of neighboring available well and seismic data values, that ensures a 0 average for the estimation error and minimizes the variance of the estimation error.

$$\begin{array}{lll} Z_L^{k^*}(x_0) & = & Z_{LTrend}^{k^*}(x_0) + \sum_{i=1}^N \lambda_i [Z(x_i) - Z_L(x_i)] \\ \text{Kriging depth estimate} & \text{Kriging of trend} & \text{Kriging weights applied to} \\ \text{of Depth of Layer L at} & \text{coefficients with Bayesian} & \text{depth residuals at Layer L} \\ \text{location } x_0 & \text{constraints} & \end{array}$$

$$\text{With } \sigma_K^2 Z_L(x_0) = \sigma_K^2 Z_{LTrend}(x_0) + \sigma_K^2 Z_{LResidual}(x_0)$$

- $\sigma_K^2 Z_L(x_0)$ uncertainty on the Layer L depth estimate , square root is depth σ map
- $\sigma_K^2 Z_{LTrend}(x_0)$ = uncertainty on the Layer L depth trend estimate, linked to uncertainty on velocity model parameters
- $\sigma_K^2 Z_{LResidual}(x_0)$ = uncertainty on the Layer L depth residual estimate, linked to the spatial covariance of time interpretation residual and velocity residuals

Using Bayesian kriging algorithm in the depth conversion workflow has the unique advantage of consistently combining the prior knowledge on the type of interval velocity model and acceptable range of parameter values with other uncertainty sources related to the time and the well data. All sources of uncertainty on velocity and time are integrated in a unique probabilistic model used for estimation or simulation of velocity and depth.

The same multivariate depth conversion Bayesian probabilistic model applies where the depth of Layer L is defined as:

$$Z_L(x) \approx \underbrace{\sum_{i=1}^L \bar{V}_i(x) \cdot \Delta t_i(x)}_{\text{Trend}} + \underbrace{\sum_{i=1}^L R_i^v(x) \cdot \Delta t_i(x)}_{\text{residual due to the velocity uncertainty}} + \underbrace{\bar{V}_L(x) \cdot R_L^t(x)}_{\text{residual due to the time uncertainty}}$$

Associated with its spatial covariance:

- $Cov(B_b, B_k) = \sigma_{lk}$ (covariance matrix for the Bayesian coefficients of the trend)
- $Cov(R_L^{v,t}(x), R_L^{v,t}(y))$ Covariance function of the depth residuals

The depth kriging estimates and Kriging variance at layer L are:

- $Z_L^{k^*}(x_0) = Z_{LTrend}^{k^*}(x_0) + \sum_{i=1}^N \lambda_i [Z(x_i) - Z_L(x_i)]$
- $\sigma_K^2 Z_L(x_0) = \sigma_K^2 Z_{LTrend}(x_0) + \sigma_K^2 Z_{LResidual}(x_0)$

Conditional simulation algorithms (turning bands, SGS ...) will compute alternative depth realizations of $Z_{(L)}^{(k^*)}(x_0)$ honoring the same multilayer trend residual covariance function and depth marker values.

The main difference compared to other common stochastic simulations consists in the fact that by construction, the Bayesian co-simulation algorithm handles both the uncertainty on the interval velocity trend and residuals instead of uncertainty on the residuals only. Simulating depth residuals around a deterministic depth base case—taken as fixed—even when using stochastic algorithms, is a major source of underestimating spatial uncertainty. Uncertainty in the velocity trend becomes dominant away from well control and has a significant impact on volumetric estimations. This directly affects closure probability maps

and the derivation of P-10, P-50, and P-90 depth scenarios, which are critical for geo-modelers to evaluate and communicate subsurface risk.

Case Study

A practical implementation of the aforementioned methodology was undertaken to ensure precise quantification of structural uncertainties within the carbonate reservoirs of Field-A. This process involved a comprehensive evaluation of the probabilistic depth conversion framework, with careful consideration given to the spatial covariance of both velocity trends and residuals. By integrating advanced Bayesian co-simulation algorithms, the analysis captured the full spectrum of uncertainties arising from velocity modeling and seismic interpretation. Multiple alternative realizations of depth scenarios were generated, thereby enabling robust volumetric assessments and more reliable subsurface risk evaluations. This rigorous approach provided critical insights into the structural complexity and uncertainty inherent to Field-A's geologic setting.

Field Overview & Accessible Datasets

Carbonate reservoirs are located on the Arabian Platform, east of the Arabian Shield, an area characterized by episodic subsidence throughout the Phanerozoic era, resulting in a substantial accumulation of sedimentary rocks from the Cambrian to the Quaternary periods. Structurally, Field-A is defined by a broad, doubly plunging anticline trending northeast-southwest, which is distinguished by a four-way closure. This structure is intersected by systematic sets of linear, small-offset conjugate strike-slip fault zones oriented approximately N75°W and N45°W, crossing the anticline at high angles relative to the axial traces. The uniform orientation and shear sense of these fault zones are consistent with development during regional WNW-ESE compressional forces, which have also been associated with Late Cretaceous inversion and subsequent fold formation.

The Field-A stratigraphic column is illustrated in Figure-2, which shows the main lithostratigraphic units and their chronostratigraphic correlation. The field's hydrocarbon reservoirs belong to the Thamama Group, a regional carbonate platform that covers most of the UAE and Oman. The Thamama Group is subdivided into several formations, based on their lithology, depositional environment, and diagenetic history. The field's reservoirs are mainly located in the Shuaiba, Kharaib, and Lekhwair formations, which correspond to the Aptian-Albian stages of the Early Cretaceous. The reservoirs are characterized by clean, porous, and fractured limestones, with varying degrees of dolomitization, cementation, and stylolitization.

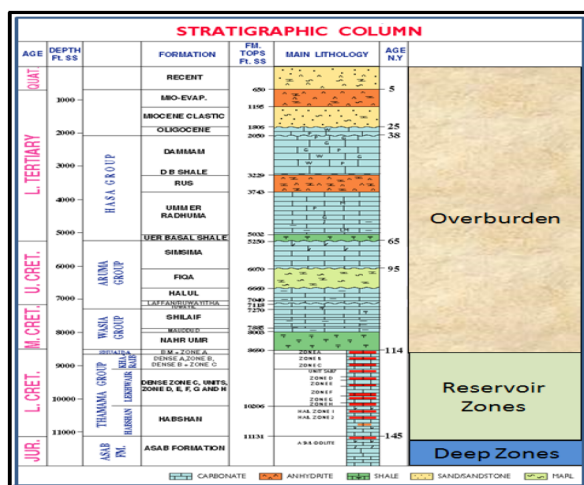


Figure 2—Generalized stratigraphy of the field

The 3D seismic data acquired in this field exhibits relatively low fold. Such restricted trace redundancy inherently diminishes signal strength and compromises the clarity of subsurface imaging, as reduced coverage heightens susceptibility to random noise and adversely affects the resolution and continuity of reflection events. The resulting low signal-to-noise ratio presents significant challenges in distinguishing authentic geological reflectors from artifacts or background noise. Furthermore, geological complexities exacerbate seismic imaging difficulties. Shallow limestone channels promote scattering and refractive effects, further complicating the interpretation of subsurface features. Above the limestone reservoirs, thick shale formations impede seismic energy transmission, particularly attenuating the high-frequency components essential for resolving detailed reflector geometries within the reservoir zones. Reservoir units introduce additional challenges to seismic imaging due to their predominantly homogeneous lithology and subtle contrasts in acoustic impedance, a function of rock density and seismic velocity, resulting in subdued reflection amplitudes and diminishing the reliability of seismic correlations. The presence of multiples within the reservoir intervals increases the complexity of the dataset. Additionally, the limited number of well control points further constrain the accuracy of velocity modeling between wells, particularly in peripheral and frontier regions of the field, thereby amplifying uncertainty in subsurface characterization.

The process of converting interpreted seismic data from time to depth domain is essential for developing accurate structural models and quantifying volumetric uncertainties in reservoirs. This step, however, introduces structural uncertainty due to variability in horizon interpretation and limitations in velocity modeling, further affected by factors such as well correlations and data quality. In this study, structural uncertainties were first systematically assessed by analyzing the variability in horizon picking and evaluating multiple velocity modeling scenarios in a classical deterministic approach. It showed that the principal contributor to structural uncertainty in the field was the variation in seismic horizon interpretation, which is critical for defining reservoir geometry. This inconsistency arose from differences in seismic data quality, signal-to-noise ratio, and interpretation methodologies. A range of alternative horizon interpretations were carried out using both PSTM and PSDM seismic datasets, with additional support from various seismic attributes to refine subsurface imaging. These efforts resulted in multiple time interpretation scenarios that capture the spectrum of potential horizon positions, summarized in Table-1, to thoroughly account for structural uncertainty in the field.

Table 1—Alternative time interpretations (Left). Table-2: Alternative velocity models (Right).

Sets	Description	Data Inputs	Remarks
Set-1	Horizon interpretation on 3D seismic data alongwith seismic attributes.	PSDM (Time), Angle Stacks, Relative Impedance & Hilbert Transform Attributes.	Best Technical Case
Set-2	Horizon interpretation on 3D seismic data. Picked the upper bound of the event to estimate the pick uncertainty.	PSDM (Time)	Alternative Interpretation
Set-3	Horizon interpretation on 3D seismic data. Picked the lower bound of the event to estimate the pick uncertainty.	PSDM (Time)	Alternative Interpretation
Set-4	Horizon interpretation on 3D seismic data.	PSDM (Time)	Alternative Interpretation
Set-5	Horizon interpretation on 3D seismic data alongwith seismic attributes.	PSTM, Angle Stacks, Relative Impedance & Hilbert Transform Attributes.	Alternative Interpretation
Set-6	Horizon interpretation on 3D seismic data alongwith seismic attributes.	PSDM (Time), Angle Stacks, Relative Impedance & Hilbert Transform Attributes.	Alternative Interpretation

Table-1: Alternative Time Interpretations

S.No.	Description	Velocities	Data Inputs	Remarks
1	Average Velocity Models	Well velocities	VSP / Checkshots & Sonic Velocities	Alternative Case
2		Well velocities with seismic velocity trends	VSP / Checkshots, Sonic & Migration Velocities	Alternative Case
3		User Defined Velocity Function	VSP / Checkshots & Sonic Velocities	Alternative Case
4		3D Grid based velocity Function	VSP / Checkshots, Sonic & Migration Velocities	Alternative Case
5	Interval Velocity Models With Overburden	Well velocities (WCS)	VSP / Checkshots & Sonic Velocities	Alternative Case
6		Well velocities (TDR)	VSP / Checkshots, Sonic Velocities	Alternative Case
7		Seismic interval velocities	Migration velocities	Alternative Case
8	Combined Velocity Models	Well velocities	VSP / Checkshots & Sonic Velocities	Alternative Case
9		Well velocities with seismic velocity trends	VSP / Checkshots, Sonic & Migration Velocities	Best Technical Case
10		3D Grid based velocity Function	VSP / Checkshots, Sonic & Migration Velocities	Alternative Case
11	Lookup Tables (TD Functions)	Well velocities	VSP / Checkshots & Sonic Velocities	Alternative Case
12	PSDM Depth Seismic	Seismic interval velocities	Migration velocities	Alternative Case

Table-2: Alternative Velocity Models

A key aspect of the deterministic structural uncertainty analysis involved compiling alternative velocity models developed for the field. Multiple velocity models, generated from sonic logs, check-shot surveys, VSP data, and seismic migration velocities, were used to assess structural uncertainty in Field-A. By

integrating well-based and seismic-derived velocity datasets, comprehensive velocity models were created for depth conversion. The corresponding layer cake velocity models are presented in Table–2. It should be emphasized that this methodology is suitable for subsurface environments characterized by gentle structural dips and minimal lateral velocity variation, thereby facilitating reliable structural interpretation of the field.

Probabilistic Structural Geomodelling Workflow

The probabilistic structural geomodelling workflow is structured into three principal steps, each designed to systematically quantify and propagate spatial uncertainties from initial data inputs through to final depth scenario outputs, as outlined in Figure-3 (*Reference: Benefits of Earth Intelligence® when modeling O&G reservoirs: An exploration case study L. Sandjiv 2023*). Machine learning algorithms are used in combination with seismic and well data to create modelled depth standard deviation maps. First, a range of machine learning methods develops a probabilistic model, often called "Generative AI", and establishes its parameters. Subsequently, an additional suite utilizes predictive methodologies such as kriging and geostatistical modelling within the structural geological framework; this phase is generally classified as unsupervised AI since it functions based on the operational framework parameters established by the first set of algorithms.

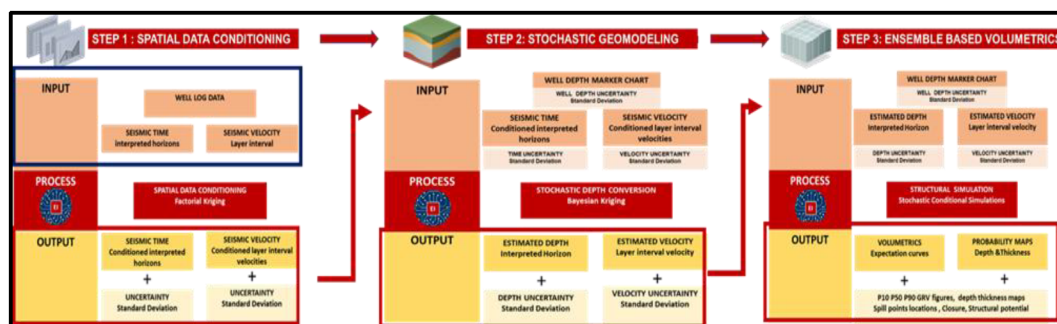


Figure 3—Steps in an intelligent structural modeling workflow, integrated within a comprehensive probabilistic framework.

Machine learning-based kriging algorithms are implemented at each step to optimize the parameters of geophysical processes, including data conditioning, velocity modeling, depth conversion, volumetric estimation, and the creation of P-10, P-50, and P-90 structural cases. Within this context, the P-50 depth scenario provides the most reliable estimate of actual structural depth, whereas the P-10 and P-90 depth scenarios are derived from the minimized modeled depth standard deviation map. This approach allows process parameters to be set without manual input, with outputs from each step being generated automatically and used as inputs for subsequent steps in the workflow.

Step 1: Data Spatial Conditioning. Spatial Data Conditioning (SDC) enhances the utility of geophysical datasets by rigorously extracting and preserving critical spatial information prior to subsequent geophysical analyses. Unlike conventional deterministic noise filtering, which often struggles to fully capture data complexity, SDC offers a robust, data-driven methodology. Utilizing advanced kriging algorithms, it optimally estimates and distinguishes the spatial components associated with signal and noise. This process is complemented by quantitative uncertainty assessments, such as kriging standard deviation maps and spatial covariance ranges, which provide spatial confidence metrics for both signal and noise characterization.

The probabilistic workflow initiates with the quantification of uncertainties inherent in input datasets, encompassing ‘Best Technical Case’ time horizon interpretations, seismic velocity fields, and well depth markers. Each of these inputs is subject to uncertainties that may impact the dependability of resultant depth maps. For instance, seismic horizon time interpretations are affected by variables including resolution, signal-to-noise ratio, and interpreter subjectivity, thereby introducing potential errors. Seismic velocity

fields, computed either from migration velocities or direct well velocities, may not precisely capture true interval velocities or lateral subsurface variations. Furthermore, well marker databases provide discrete depth measurements that can be influenced by factors such as borehole deviation, instrument inaccuracies, or datum corrections. It should be emphasized that the probabilistic workflow requires only one set of seismic interpretations from the latest reprocessing as input, differing from deterministic workflows, which typically need multiple alternative interpretations. The full well marker database is used, except for a subset of blind wells reserved for independent validation of probabilistic depth conversion accuracy in selected structural regions.

The Factorial Kriging (FK) workflow is applied to horizon time interpretations. These time interpretations are decomposed into two distinct components: a spatial trend, representing low spatial frequency variations, and a residual component, capturing mid-to-high spatial frequency features. This methodology employs RMS amplitudes and dominant frequency values to calculate a stacked amplitude attribute known as "relative contrast," as described by Ramos Pinto et al. (2017).

$$A_{rms}^n = 1 + \left(1 - \frac{A_{rms}}{\max(A_{rms})}\right) \cdot p$$

The representation of spatial uncertainty within time interpretation leverages a spatial covariance or variogram function. In this context, the 'sill' parameter is estimated based on the methodology proposed by Ramos Pinto et al., while the range is derived from the variogram map constructed using the time residual data.

As detailed above, when evaluating seismic velocities, the factorial kriging model decomposes an interval PSDM velocity field into two primary components: a "geological" spatial element and a local residual element representing geophysical artifacts. The residual component lacks correlation with the time isopach and is therefore not geologically meaningful. The geological component is determined via kriging, and its associated standard deviation offers a rigorous quantitative assessment of spatial uncertainty. This methodology enables a systematic evaluation of confidence in the extracted structural signal. FK workflow can be used on the well-marker data to identify outliers and assign uncertainty values, allowing for more accurate weighting in modeling. In this case, the well-marker dataset was already consistent and matched the one used in the deterministic model, so no preliminary adjustments were needed.

The spatial conditioning phase yields a suite of outputs for seismic horizon time interpretations, including the original seismic time picks, spatial uncertainty maps representing the first standard deviation (1σ), and corresponding spatial correlation ranges. For pre-stack depth migration (PSDM) layer interval velocities, the outputs comprise the conditioned interval velocity map, an associated spatial uncertainty map (1σ), and the spatial correlation range utilized for quantifying time-related uncertainties. [Figure-4](#) & [Figure-5](#) illustrate the deliverables generated for Reservoir-A. Collectively, these deliverables provide a robust quantitative basis for subsequent modeling processes.

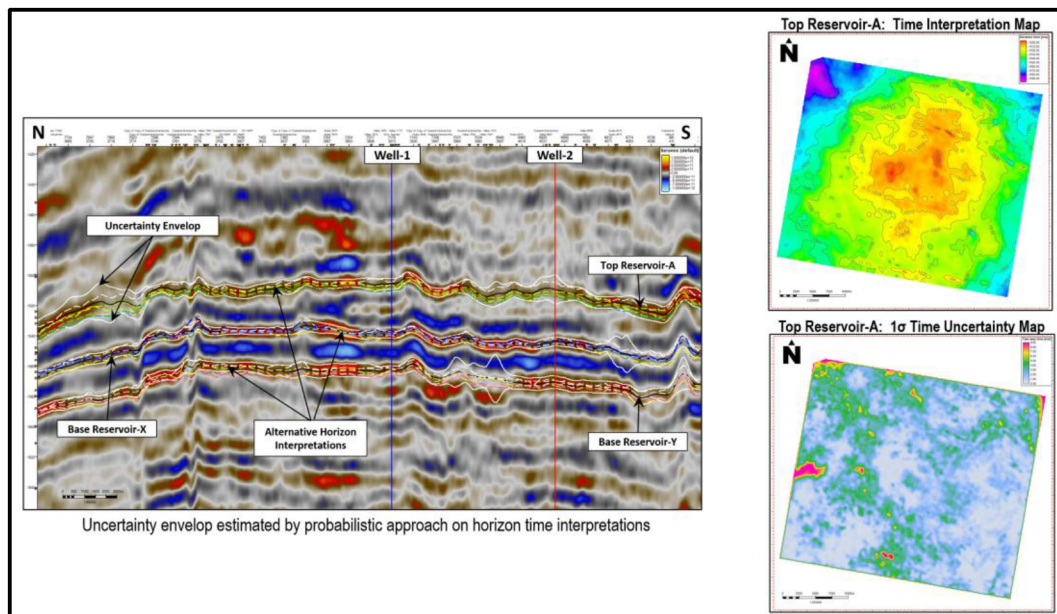


Figure 4—Overview of key deliverables resulting from spatial data conditioning based on horizon interpretations. The range of alternative horizon picks falls within the uncertainty bounds established by the probabilistic methodology, thereby validating the robustness of this estimation approach.

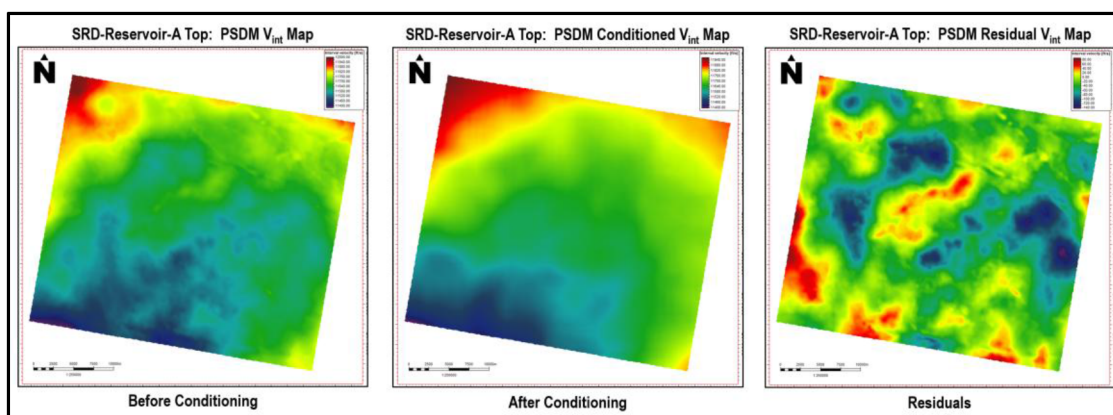


Figure 5—Outcomes after velocity conditioning, as produced during the spatial data conditioning phase.

Step 2: Velocity Model Construction and Horizon Depth Conversion. Probabilistic AI seismic-driven geomodelling, also referred to as stochastic geophysical modeling (SGM), enhances the accuracy of subsurface property models by integrating geophysical data with well data, while explicitly accounting for uncertainty. In contrast to traditional approaches that predominantly extrapolate from well data, SGM directly estimates properties utilizing geophysical inputs, with well information providing additional guidance. The Multilayer Bayesian Co-Kriging algorithm employs a probabilistic framework that incorporates spatial uncertainty (σ) maps derived from time and velocity data. This methodology enables comprehensive uncertainty propagation, generating depth spatial uncertainty maps for each depth-converted horizon, thereby capturing both the spatial variability and the reliability of the modeled geological structures.

The input for velocity modeling and depth conversion should comprehensively incorporate all pertinent horizons to be depth converted. The Multivariate Bayesian cokriging algorithm determines layer depths based on depth marker charts, applying predefined velocity parameters for each individual layer. Within probabilistic geomodelling frameworks, interval velocity models are derived by converting time-interpreted horizons into depth. Conversely, deterministic methodologies construct interval velocity models through

the sequential allocation of velocities to layers prior to depth conversion, with subsequent adjustments made to align with well markers as necessary.

Utilizing a series of predefined velocity laws assigned to each layer, the Multivariate Bayesian Co-Kriging algorithm runs on data well locations only and estimates the optimal (posterior) values of all velocity law parameters in a single run. These posterior values are constrained to remain within the predefined "a priori" ranges provided as input. The algorithm also evaluates the performance of the selected velocity laws by analyzing the depth mistie statistics at well locations, thus providing a quantitative measure of how accurately the model incorporates well data. Following the verification of layer velocity assumptions through depth mistie statistics at well locations, i.e., the posterior values yield an average depth mistie close to zero and a modeled standard deviation that is comparable to or exceeds the experimental standard deviation, the Multivariate Bayesian Co-Kriging is applied across the entire grid covering the seismic area.

The velocity model building and depth conversion process using Multivariate Bayesian Co-Kriging produces a probabilistic velocity model that quantitatively characterizes both uncertainty and data fit, alongside a corresponding probabilistic depth map that reflects associated uncertainties and accuracy. This depth map is generated by integrating the velocity model with the time-based interpretation of seismic horizons through a depth conversion algorithm that rigorously accounts for uncertainty in both velocity and temporal domains. The product delivers depth estimates for each seismic horizon, accompanied by quantified uncertainties such as standard deviation and confidence intervals (refer to Figure-6). Interval velocities for each layer are determined with corresponding uncertainty assessments (see Figure-7). Figure-8 presents a cross-section from the 3D velocity model following velocity modeling and depth conversion.

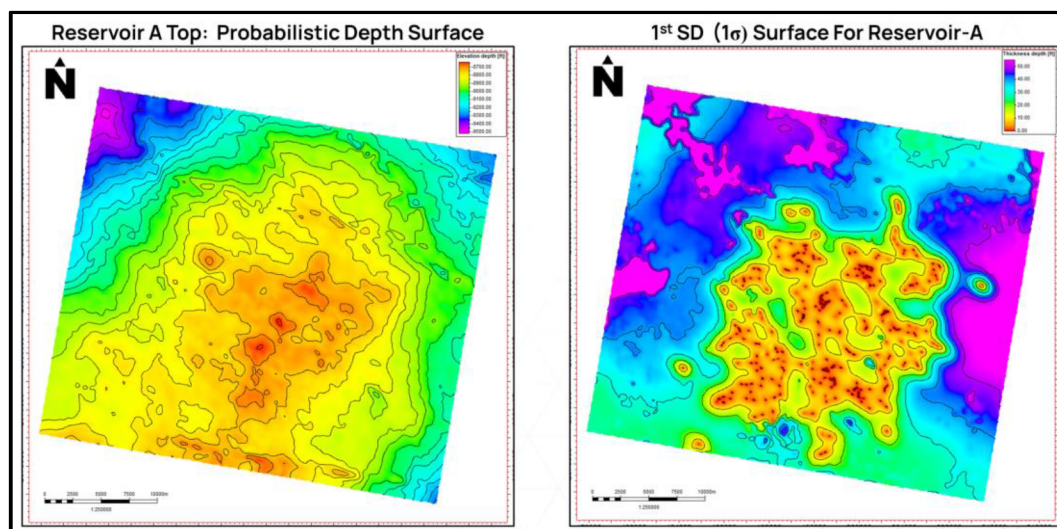


Figure 6—The optimal seismic horizon depth estimation and its associated depth uncertainty, as quantified by the 1st standard deviation (1σ) from the velocity modeling and depth conversion process.

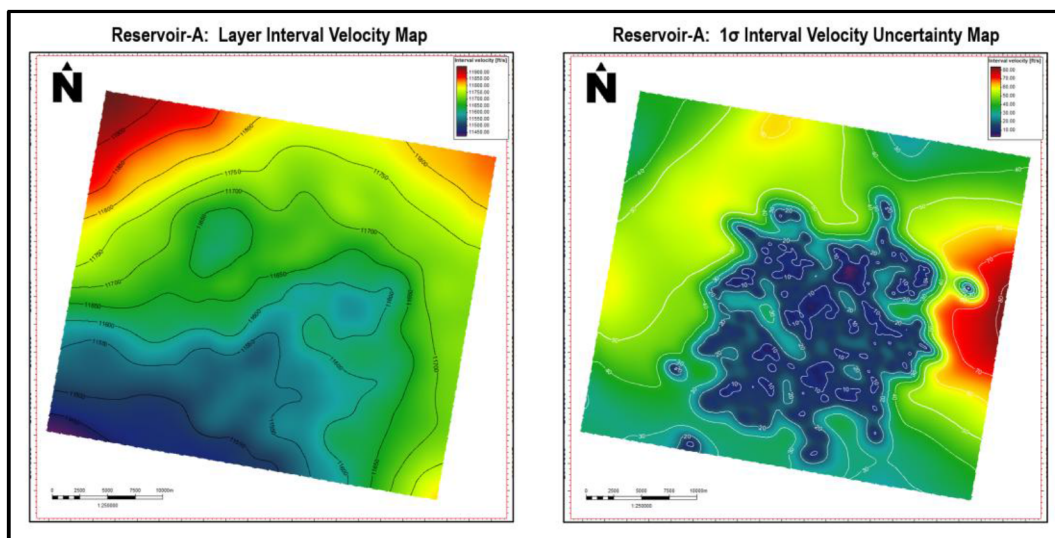


Figure 7—The layer interval velocity estimation and its associated velocity uncertainty, as quantified by the 1st standard deviation (1σ) from the velocity modeling and depth conversion process.

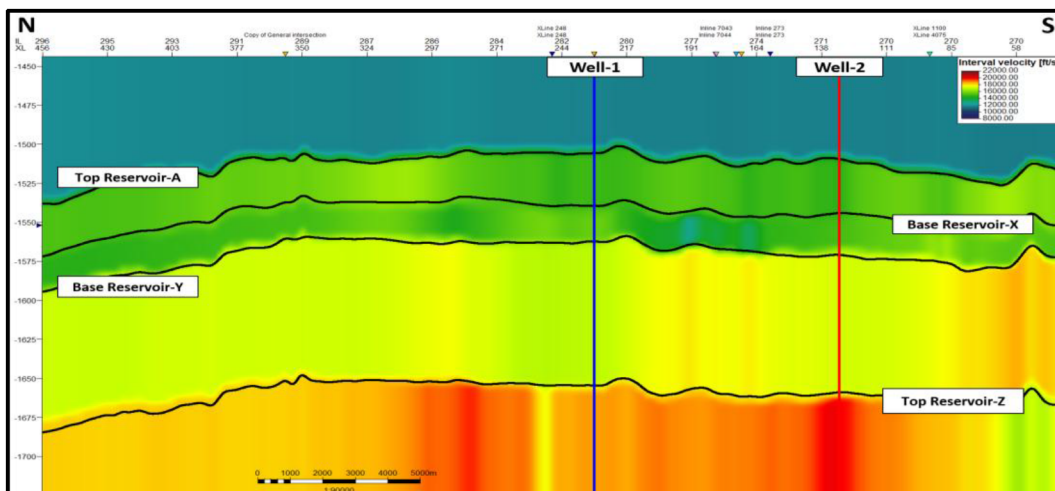


Figure 8—N-S Cross section from the 3D velocity model output from the velocity modeling and depth conversion process.

Step 3: Generation of Volumetric P-10, P-50, and P-90 Depth Scenarios. Application of an oil-water contact (OWC) cut-off to a kriged depth map or deterministic base case does not yield reliable volumetric, closure, or spill point calculations, owing to the inherent non-linearity between volume above the cut-off and depth. Such methods are prone to introducing systematic and unpredictable biases. A more rigorous approach leverages probabilistic modeling, employing the same multilayer covariance framework utilized in depth conversion. This methodology enables simulation of the spatial distribution of the unknown "true" depth, constrained by both the covariance structure and available well markers. By generating hundreds or thousands of equiprobable depth realizations, a consistent training set is established for advanced AI algorithms. These algorithms can then account for the complex, non-linear behavior of depth, ensuring unbiased estimations of volumetrics, structural closures, and spill point positions.

To maintain alignment with the comprehensive uncertainty quantification framework, the generation of simulated depth maps should extend beyond reliance on the standard depth sigma map typically used by geo-modelers. Conditional simulations must incorporate the same covariance model applied during depth conversion, thereby propagating spatial uncertainty both as deviations around the deterministic depth base case and within the base case itself via simulation of velocity trend parameters. This approach ensures

that the resulting ensemble of depth realizations captures the complete spectrum of structural uncertainty, ultimately enhancing the reliability of volumetric assessments and structural interpretations.

Probabilistic co-simulation of depth horizons is conducted within the same Bayesian multilayer framework as Step 2. The multivariate probabilistic model utilized in the depth conversion process is further applied to generate 200 depth realizations that rigorously explore the full range of spatial uncertainty, encompassing both trend uncertainty and residual deviations. These simulations are constrained by both static (geological) and dynamic (reservoir) criteria to produce credible volumetric estimates and P10, P50, and P90 structural scenarios, effectively capturing spatial variability and modeling uncertainty.

A comprehensive suite of depth scenarios generated through simulation was subsequently utilized to compute the Gross Rock Volume (GRV) above the Free Water Level (FWL) for each reservoir. The GRV values derived from 200 realizations within the modeling scenarios exhibit a distribution that approximates normality, as evidenced by the close alignment of mean and median values and the bell-shaped appearance of the corresponding histograms. Consequently, the predicted GRV for each scenario effectively represents the range of variability attributable to structural uncertainty in the initial best-estimate structure. Figure-9 presents histograms and Cumulative Distribution Functions (CDFs), depicting the P-10, P-50, and P-90 scenarios for Reservoir-X.

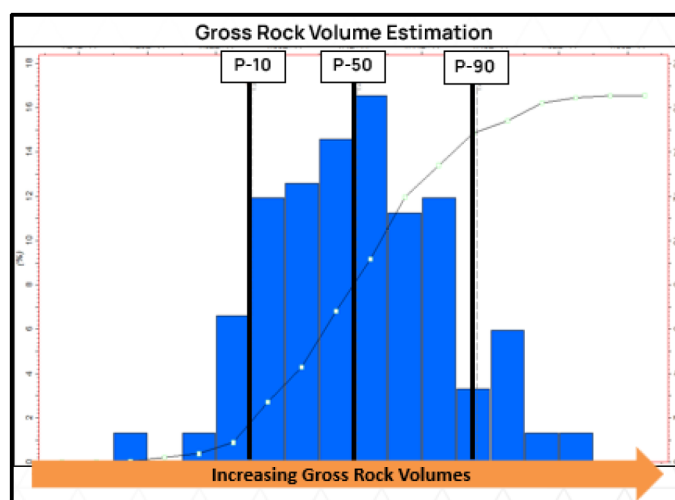


Figure 9—Histograms & CDFs for GRVs containing the P10, P50, P90 scenarios for Reservoir-X.

A comprehensive understanding of reservoir geometry and connectivity is fundamental to successful oil field development, as these characteristics dictate hydrocarbon distribution and recovery efficiency. Critical attributes such as reservoir size, orientation, stratigraphic arrangement, and fluid contact levels play a pivotal role in guiding optimal well placement and informing robust reservoir management strategies. The thorough characterization of these features is therefore essential for effective field development and the maximization of resource recovery. Leveraging stochastic simulations of the structural model has significantly advanced our understanding of reservoir geometry and connectivity. This approach enables the generation of P-10, P-50, and P-90 depth scenarios, as well as the development of comprehensive reservoir thickness and probability maps, and the delineation of spill point locations. The results of these simulations for Reservoir X are presented in Figure-10.

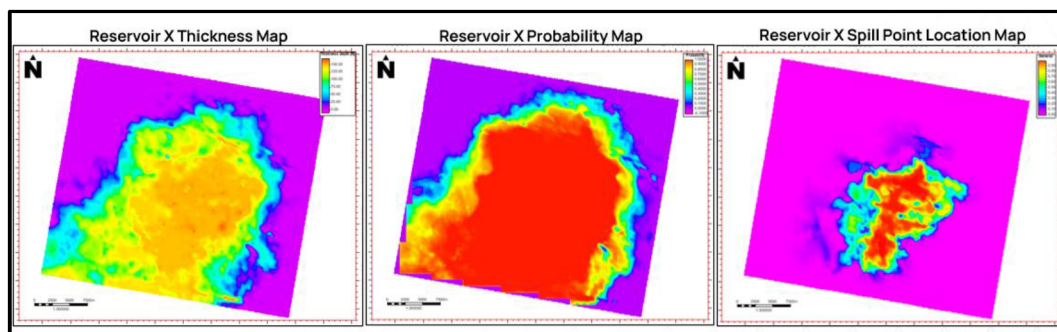


Figure 10—Reservoir thickness and probability maps estimate reservoir geometry, including shape, size, orientation, and fluid contact. Spill point maps help understand the connectivity of reservoir units that impact hydrocarbon flow and extraction.

Results & Analysis

Determining the optimal level for residual uncertainties is a nuanced process, requiring iterative refinement and a deep understanding of the data landscape. The magnitude of these uncertainties directly influences the simulated horizons and, by extension, the Gross Rock Volume (GRV) estimations. By proactively capturing uncertainty envelopes during seismic interpretation, we ensure that these variabilities are consistently and transparently represented right from the data's inception. This approach manifests in the uncertainty map (1σ), which quantifies depth errors and provides a reliable visual summary of the uncertainty analysis.

Effective data validation is pivotal, forming the backbone of robust quality assurance as advanced methodologies transition from conceptual innovation to actionable insights. Benchmarking outputs from probabilistic, AI-driven frameworks against those from traditional deterministic analyses allows practitioners to harness the distinct strengths of each while addressing their respective limitations. This disciplined comparison not only safeguards data integrity but also fosters transparency and continual enhancement within increasingly complex modeling environments.

Probabilistic depth conversion methodologies integrate uncertainties associated with velocity models and seismic data to produce depth estimates accompanied by rigorously quantified confidence intervals. In contrast, deterministic approaches assign velocities to stratigraphic layers in a sequential manner prior to depth conversion, with subsequent adjustments made to honor well markers. A comparative overview of the resulting depth estimates from both methodologies is presented in Figure-11.

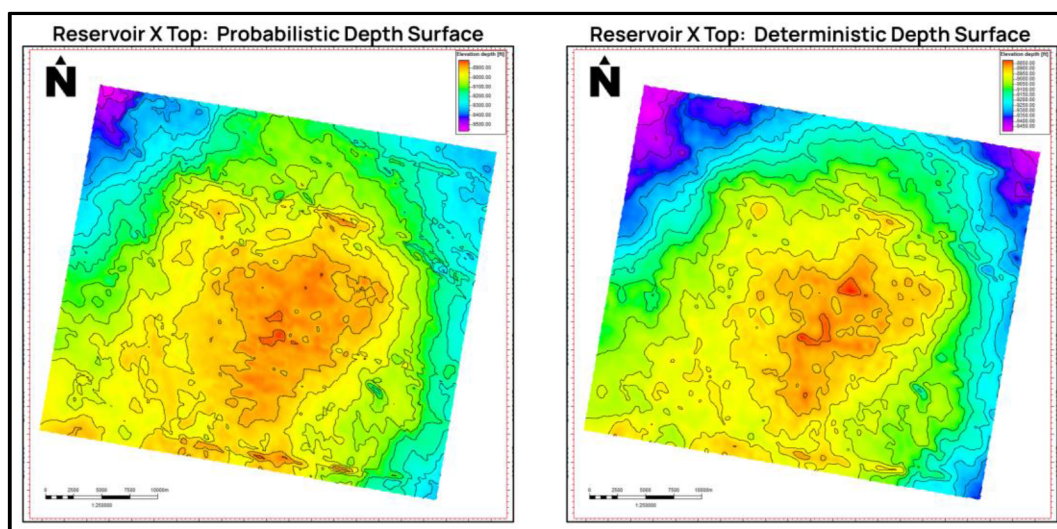


Figure 11—Comparison of the depth surfaces, for Reservoir-X Top, generated using the probabilistic approach versus the deterministic approach as part of the results validation.

In this context, standard deviation maps of depth error (1σ), as shown in Figure-12, demonstrate that AI-supported probabilistic depth conversion methods provide a more comprehensive assessment of uncertainty compared to deterministic approaches. By accounting for spatial variability and the uncertainties inherent in seismic data, velocity models, and well observations, these techniques produce structural models and depth predictions with enhanced reliability. The deterministic results further validate the robustness of AI-assisted probabilistic workflows. Both methodologies consistently demonstrate reduced depth errors in regions with strong well coverage, whereas increased uncertainties are observed in areas with limited well picks, which is consistent with established theoretical models.

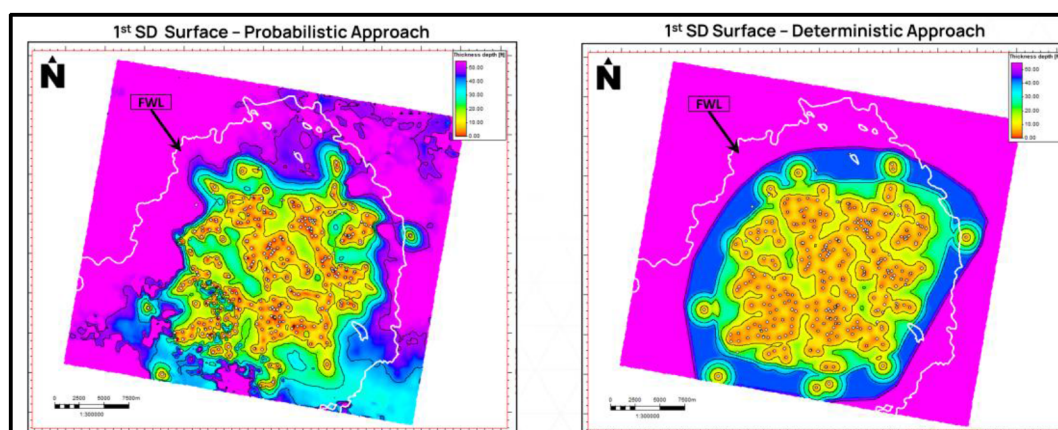


Figure 12—Comparison of the 1st standard deviation depth error surfaces, for Reservoir-X, generated using the probabilistic approach versus the deterministic approach as part of the results validation.

Comparative analysis of Gross Rock Volume (GRV) and in-place hydrocarbon estimations across multiple depth scenarios underscores the high degree of alignment between both deterministic and AI-driven probabilistic methodologies: GRV discrepancies are limited to just 0.5% (P-10), 0.2% (P-50), and 0.7% (P-90), as detailed in Figure-13. These minimal differences demonstrate the robustness of advanced uncertainty quantification and the precision afforded by AI-enhanced depth prediction workflows, which thoroughly integrate spatial and inherent data uncertainties from seismic, velocity, and well sources. The corroboration provided by traditional deterministic methods further reinforces reliability and confidence in the integrated probabilistic approach.

This consistent narrative illustrates that integrating AI-enabled probabilistic modeling with established deterministic workflows enhances the understanding of subsurface structures, supporting informed decision-making and optimizing value throughout all phases of reservoir development.

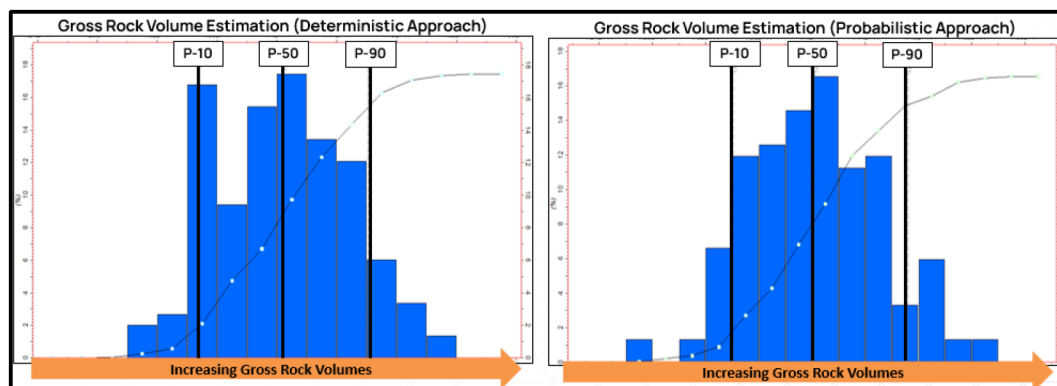


Figure 13—Histograms and cumulative distribution functions (CDFs) for gross reservoir volumes (GRVs), illustrating the P10, P50, and P90 scenarios derived from both probabilistic and deterministic approaches for Reservoir-X.

Reservoir depth thickness and probability maps, as demonstrated in Figure-7, are essential tools for accurately estimating key aspects of reservoir geometry, including shape, size, orientation, and spatial configuration, as well as fluid contact levels. Integrating closure probability maps with 1σ depth error maps allows for the creation of structural potential maps (refer to Figure-14), which highlight regions characterized by both elevated uncertainty and high closure probability. Prioritizing these areas enables a more precise refinement of the P10–P90 GRV range, thereby facilitating robust and informed development strategies. Additionally, the spill point location map, also referenced in Figure-7, is instrumental in elucidating the connectivity between reservoir units, which directly influences hydrocarbon distribution, migration pathways, and extraction efficiency.

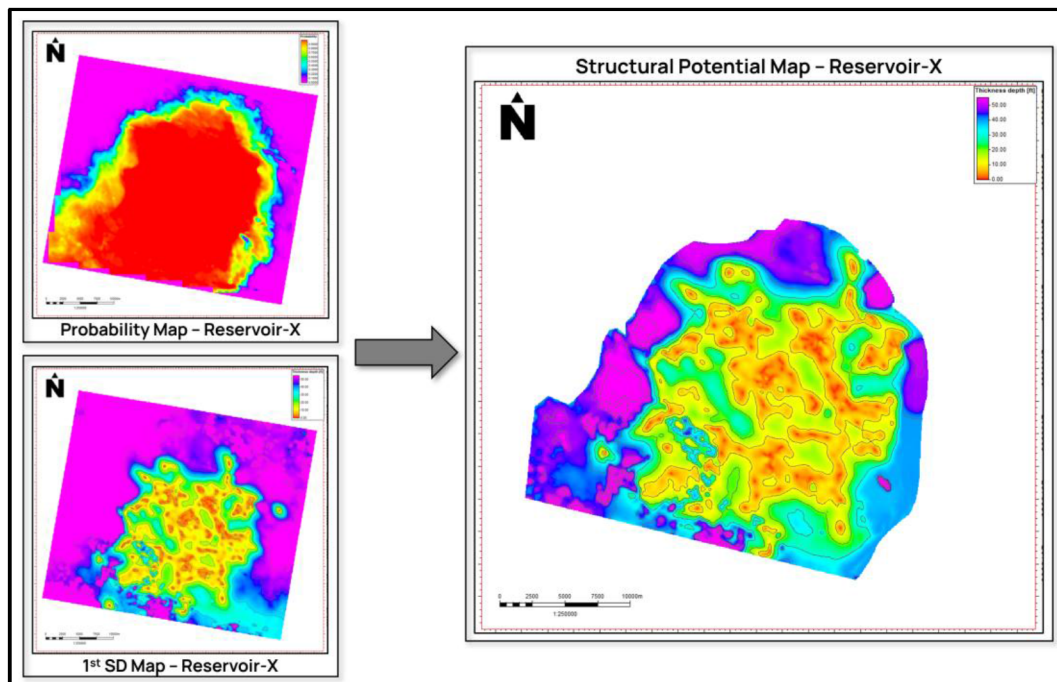


Figure 14—Structural potential map indicating the locations with both high depth uncertainty and high inside closure probability, which are identified as areas to be explored for best narrowing the P10 P90 GRV range.

These comprehensive visualizations inform the strategic placement and spacing of wells to maximize reservoir exposure, optimize drainage areas, and mitigate operational challenges such as water or gas coning. The development team leverages these insights to evaluate alternative drilling locations, thereby enhancing operational flexibility and field development planning. Furthermore, these maps underpin the

design of effective reservoir management strategies, encompassing pressure maintenance, waterflooding, and enhanced oil recovery initiatives.

By providing a data-driven foundation for well placement and management decisions, these mapping products support the maximization of resource recovery, address issues such as premature water or gas breakthrough and bypassed hydrocarbons and help mitigate risks associated with reservoir compaction. Overall, the integration of these maps into the reservoir development workflow is critical for improving recovery efficiency and ensuring the long-term viability of hydrocarbon extraction projects.

AI-driven probabilistic structural geomodelling workflows signify a paradigm shift in how subsurface depth uncertainty is analyzed and managed within hydrocarbon exploration. Throughout this technical narrative, we have outlined a progression wherein traditional deterministic approaches, while foundational, often encounter limitations due to their inability to fully capture the complex, non-linear interactions inherent to subsurface geology. This technical discussion traces the evolution from such established methods toward the integration of advanced Artificial Intelligence (AI) and probabilistic techniques, culminating in a more comprehensive and reliable uncertainty quantification process.

Central to this transformation is the ability of AI-based probabilistic frameworks to seamlessly assimilate and interpret vast, heterogeneous datasets, ranging from seismic surveys and velocity models to well logs and ancillary geological data. Unlike deterministic analyses that typically rely on fixed parameter sets and single-scenario outputs, these AI-enhanced methodologies leverage sophisticated algorithms to map intricate relationships and propagate uncertainty across the entire modeling workflow. This approach ensures that both spatial and inherent data uncertainties are fully accounted for, resulting in more robust and realistic predictions of subsurface structures.

As highlighted through the technical narrative, the implementation of AI-driven probabilistic workflows delivers significant operational benefits. The automation of feature extraction and data integration streamlines processes that were once predominantly manual, minimizing the potential for human error and enabling technical teams to concentrate on higher-level analysis and strategic planning. These workflows are inherently scalable, effectively accommodating the large and dynamic datasets encountered in modern exploration initiatives. Moreover, their adaptive nature allows for continuous refinement as new data and operational insights emerge, ensuring models remain current and relevant. This adaptability is particularly valuable in fast-paced operational contexts, where near-real-time updates and decision support are essential for maintaining a competitive edge. Moreover, the incorporation of anomaly detection within AI workflows significantly enhances data quality and reliability. By systematically identifying and addressing outliers and inconsistencies, these systems facilitate timely data corrections, thereby strengthening the credibility of model outputs. This heightened assurance in structural and depth predictions is essential for informed decision-making related to well placement, drilling strategies, and comprehensive field development planning.

The narrative also emphasizes the collaborative synergy between AI-enabled probabilistic approaches and legacy deterministic methods. Rather than supplanting traditional workflows, AI acts as an augmentative layer, validating and cross-referencing outputs to ensure consistency and transparency. This dual-framework strategy not only supports greater technical assurance but also facilitates organizational buy-in, as stakeholders can evaluate results from both familiar and novel perspectives. The implementation of these advanced workflows has led to significant operational enhancements. Projects have reported reductions in manual workload and operational costs, with turnaround times for complex depth uncertainty analysis shrinking from several months to mere weeks. Most notably, the move to probabilistic, data-driven forecasts, eschewing reliance on single-point estimates, enables more nuanced risk assessments and supports robust, evidence-based decision-making throughout the hydrocarbon project lifecycle. In summary, the storyline illustrates that the integration of AI-driven probabilistic modeling into established deterministic frameworks delivers superior predictive accuracy, operational efficiency, and flexibility. This synergy fundamentally elevates the standards of subsurface characterization, empowering practitioners to

manage uncertainty more effectively and optimize value across all phases of hydrocarbon exploration and production.

Conclusions

The integration of Artificial Intelligence (AI) and advanced probabilistic methodologies has ushered in a transformative era for geoscientific modeling. This comprehensive study has detailed how AI-driven frameworks, harmonized with foundational geostatistical theories such as kriging, now provide a robust, scalable, and efficient approach to structural uncertainty quantifications surpassing the limitations of traditional deterministic modeling. A central advancement lies in the seamless combination of data-driven machine learning algorithms and rigorous statistical methods, enabling practitioners to accurately capture, propagate, and visualize uncertainty throughout the structural modeling workflow. The probabilistic geomodelling process supports the optimal estimation of subsurface properties by explicitly modeling both the mean trends and spatial covariance structures within geophysical datasets. This dual focus not only ensures unbiased and minimum-variance predictions but also provides confidence intervals and scenario-based outputs essential for risk-aware decision-making.

AI-enhanced workflows have demonstrated a marked improvement in automating intricate and time-consuming tasks, dramatically reducing project turnaround times, from several months to mere weeks, while maintaining high standards of accuracy and reproducibility. The adoption of advanced kriging algorithms and conditional spatial simulations empowers asset teams to generate multiple equiprobable realizations of subsurface structures. These realizations not only capture the full spectrum of spatial uncertainty but also serve as valuable training sets for further AI and machine learning model development. A significant operational benefit of these methodologies is their capacity to integrate diverse and heterogeneous data sources, including seismic surveys, well markers, and velocity models, into a unified probabilistic framework. This integration ensures that uncertainties originating from data quality, acquisition methods, and geological complexity are systematically accounted for and transparently communicated. The result is an improved alignment between model outputs and actual subsurface conditions, as validated through blind well testing and comparison with deterministic workflows.

Comparative analyses between deterministic and AI-assisted probabilistic approaches reveal a high degree of consistency in volumetric estimations and structural predictions, with discrepancies in key metrics such as Gross Rock Volume (GRV) remaining below 1% across P-10, P-50, and P-90 scenarios. This close agreement both reinforces confidence in the new methods and highlights their enhanced ability to capture and communicate the multi-dimensional nature of uncertainty. Moreover, the professional adoption of these workflows supports a paradigm shift from reliance on single-point estimates to a culture of comprehensive, data-driven uncertainty management. This transition enables more nuanced reservoir management strategies, optimizes well placement, and facilitates more informed investment decisions. The adaptability and scalability of AI-driven probabilistic models further position them as indispensable tools as the industry continues to confront increasingly complex geological challenges and the demand for rapid, reliable insights grows.

In summary, the fusion of AI with probabilistic structural modeling establishes a new standard for subsurface characterization. By providing deeper, more actionable understanding of geological uncertainty, these methodologies equip geoscientists and reservoir engineers with the tools necessary to maximize resource recovery, mitigate technical and economic risks, and drive value throughout the entire lifecycle of hydrocarbon exploration and production. As the industry continues to evolve, these integrated approaches will remain at the forefront of innovation, ensuring that decisions are grounded in the most rigorous, transparent, and data-rich foundations available.

Acknowledgements

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